

Smart Employee Management System – Full Updated Team Work Division (12 Members)

Member 1 — Authentication Backend Developer

Responsibilities: authController.js, authRoutes.js, JWT Authentication, Password hashing, Role-based authentication

Tasks: User registration API, User login API, JWT token generation, Protect private routes, Authentication middleware

Backend Files: controllers/authController.js, routes/authRoutes.js, middleware/authMiddleware.js, middleware/roleMiddleware.js

Technologies: jsonwebtoken, bcryptjs

Member 2 — Login Frontend Developer

Responsibilities: Frontend login system

Tasks: Login page UI, Form validation, API integration, JWT token storage, Redirect after login, Logout functionality

Frontend Files: pages/Login.jsx, components/LoginForm.jsx, services/authService.js, context/AuthContext.jsx

Features: Remember logged-in user, Display login errors, Secure token handling

Technologies: React, Axios

Member 3 — Register Frontend & Backend Developer

Responsibilities: Registration system

Tasks: Registration form, User role selection, Password confirmation, Backend integration, Validation handling

Frontend Files: pages/Register.jsx, components/RegisterForm.jsx

Backend Files: controllers/authController.js, routes/authRoutes.js

Features: Register Admin, Register HR, Register Employee, Register Manager

Member 4 — Employee Management Developer

Responsibilities: Employee management module

Tasks: Add employees, Update employee details, Delete employees, Search/filter employees, Employee document uploads, Generate employee IDs

Backend Files: controllers/employeeController.js, routes/employeeRoutes.js, models/Employee.js, utils/generateEmployeeId.js

Frontend Files: pages/Employees.jsx, components/EmployeeTable.jsx, components/AddEmployeeModal.jsx

Member 5 — Attendance Management Developer

Responsibilities: Attendance system

Tasks: Mark attendance, Login attendance tracking, Logout attendance tracking, Present/Late calculation, Attendance history

Backend Files: controllers/attendanceController.js, routes/attendanceRoutes.js, models/Attendance.js

Frontend Files: pages/Attendance.jsx, components/AttendanceTable.jsx

Features: Login time tracking, Logout time tracking, Daily attendance report

Member 6 — Smart Session & Activity Tracking Developer

Responsibilities: Automatic session management

Tasks: Save login start time automatically, Save logout end time automatically, Session duration calculation, Inactive employee detection, Auto logout handling

Features: startTime = new Date(), endTime = new Date(), Working Hours = End Time - Start Time

Backend Files: controllers/attendanceController.js, controllers/authController.js

Frontend Files: components/Navbar.jsx, pages/Dashboard.jsx, utils/sessionTracker.js

Member 7 — Leave Management Developer

Responsibilities: Leave management module

Tasks: Apply leave, Leave approval/rejection, Leave cancellation, Medical document uploads, Leave balance calculation

Backend Files: controllers/leaveController.js, routes/leaveRoutes.js, models/Leave.js

Frontend Files: pages/Leave.jsx, components/LeaveForm.jsx, components/LeaveHistory.jsx

Features: Multiple leave types, HR approval workflow, Manager review system

Member 8 — Payroll Management Developer

Responsibilities: Payroll management system

Tasks: Salary calculations, Tax calculations, Loan deductions, Generate payroll records, Payslip generation

Backend Files: controllers/payrollController.js, routes/payrollRoutes.js, models/Payroll.js

Frontend Files: pages/Payroll.jsx, components/Payslip.jsx

Salary Formula: $\text{Net Salary} = (\text{Basic Salary} + \text{Allowances}) - (\text{Deductions} + \text{Tax} + \text{Loans})$

Member 9 — Task Management Developer

Responsibilities: Task management module

Tasks: Assign tasks, Update task progress, Add task comments, Task status tracking

Backend Files: controllers/taskController.js, routes/taskRoutes.js, models/Task.js

Frontend Files: pages/Tasks.jsx, components/TaskBoard.jsx, components/TaskCard.jsx

Workflow: To Do → In Progress → Review → Completed

Member 10 — Performance Management Developer

Responsibilities: Performance evaluation system

Tasks: Calculate performance scores, Attendance scoring, Task completion scoring, Quality assessment, Manager feedback system

Backend Files: controllers/performanceController.js, routes/performanceRoutes.js, models/Performance.js

Frontend Files: pages/Performance.jsx, components/PerformanceChart.jsx

Performance Formula: $\text{Overall Score} = (\text{Attendance Score} + \text{Task Completion Rate} + \text{Quality Score}) / 3$

Member 11 — Frontend Dashboard & UI Developer

Responsibilities: Main frontend UI system

Tasks: Admin dashboard, Employee dashboard, HR dashboard, Sidebar and navbar, Charts and statistics, Responsive UI design, Notification UI

Frontend Files: pages/Dashboard.jsx, pages/AdminDashboard.jsx, pages/EmployeeDashboard.jsx, pages/HRDashboard.jsx, components/Sidebar.jsx, components/Navbar.jsx, components/Charts.jsx

Technologies: React, Tailwind CSS, Chart.js

Member 12 — Reports, Notifications & Documentation Developer

Responsibilities: Reports and documentation system

Tasks: Attendance reports, Payroll reports, Leave reports, Generate PDF exports, Email notifications, Payslip downloads, Project documentation, API documentation, User manual, Final presentation support

Backend Files: controllers/reportController.js, controllers/notificationController.js, services/pdfService.js

Frontend Files: pages/Reports.jsx, pages/Notifications.jsx, components/DownloadPayslip.jsx

Technologies: PDFKit, Nodemailer

